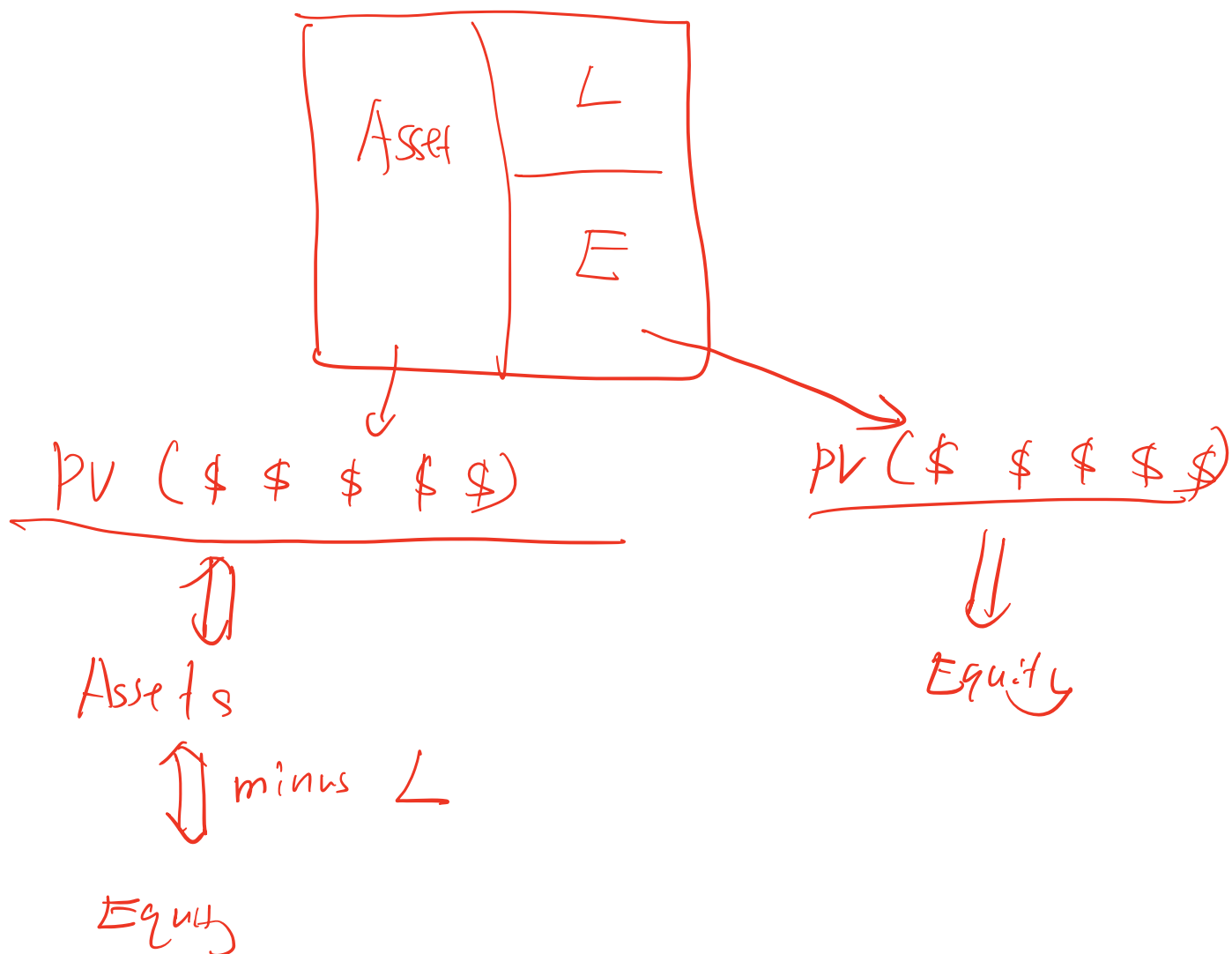


Module 3: Discounted Cash Flow (DCF) Valuation

From Target Firm's Balance Sheet to Valuation

- Assume that the acquirer aims to buy 100% of the target's equity.
- Important considerations when valuing a target firm:
 1. An acquirer purchases a target firm's business
 - thus, a starting point would be to value a target firm's **assets** (LHS of the balance sheet)
 2. There are multiple claims on cash flows from firm's assets (equity, debt, non-debt liabilities)
 - an acquirer becomes the owner of target firm's **equity**
 - thus, need to estimate the value of residual claims on firm's cash flows-the value of target equity
 - **equity** = **assets** minus the values of **debt** and **non-debt liabilities**



DCE

Begin with the Target Firm's Balance Sheet

$$\text{Net Working Capital} = \text{Current assets} - \text{Current liabilities}$$

- Cash
- Inventory
- Accounts receivables

Current Assets

- P.P.E.
- Intangible assets

Other Assets

Current Liabilities

- Accounts payable
- Accrued expenses,
- Taxes payable

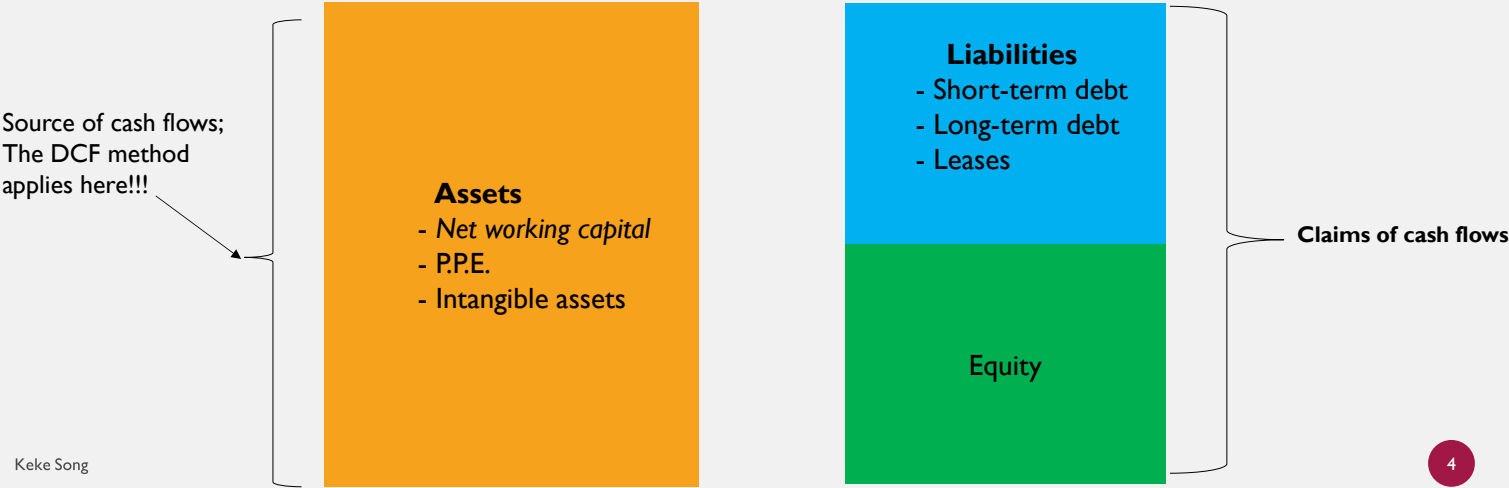
Non-Current Liabilities

- Short-term debt
- Long-term debt
- Leases

Equity

Slightly Modified Target Firm's Balance Sheet for Valuation

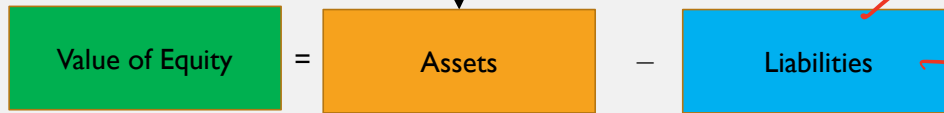
- We treat net working capital (rather than current assets) as part of assets.
- So, the liabilities only include the non-current liabilities such as debt (also called borrowing) and leases.



From Target Firm's Balance Sheet to Valuation (cont'd)

- Using the balance sheet identity:

The cash-flow-generating box & "trickiest" part in valuation!



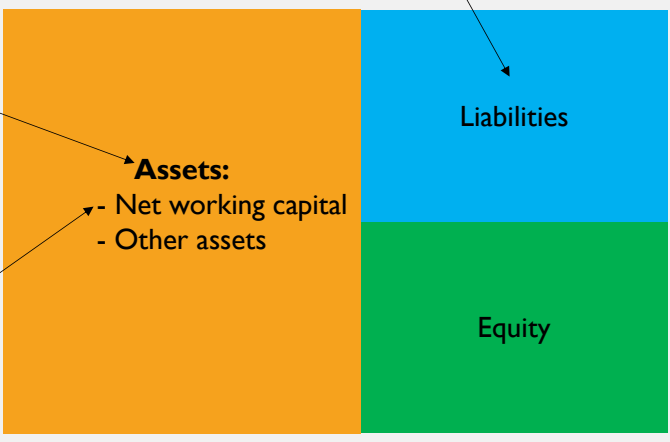
- We shall go through the estimation of each of these "valuation blocks" in turn.

Notes on the underlying assumptions of our DCF approach

3. Short-term debt belongs to NON-current liabilities.

1. All the assets are operating assets (namely, all assets are involved in generating the cash flows).

2. All cash holding (also called "cash balance") is assumed to be part of the net working capital and, therefore, part of operating assets.



Keke Song

Assets = V (Op. Assets) $\Leftrightarrow$ PV (\$ \$ \$ \$ \$)
by assumption $\$500m$ $100m.$

500
100
50

650

non-op. assets

- Vacant Land $\rightarrow \$100m$
- Excess Cash $\rightarrow \$50m$

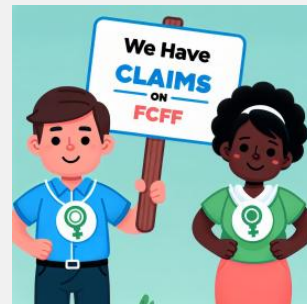
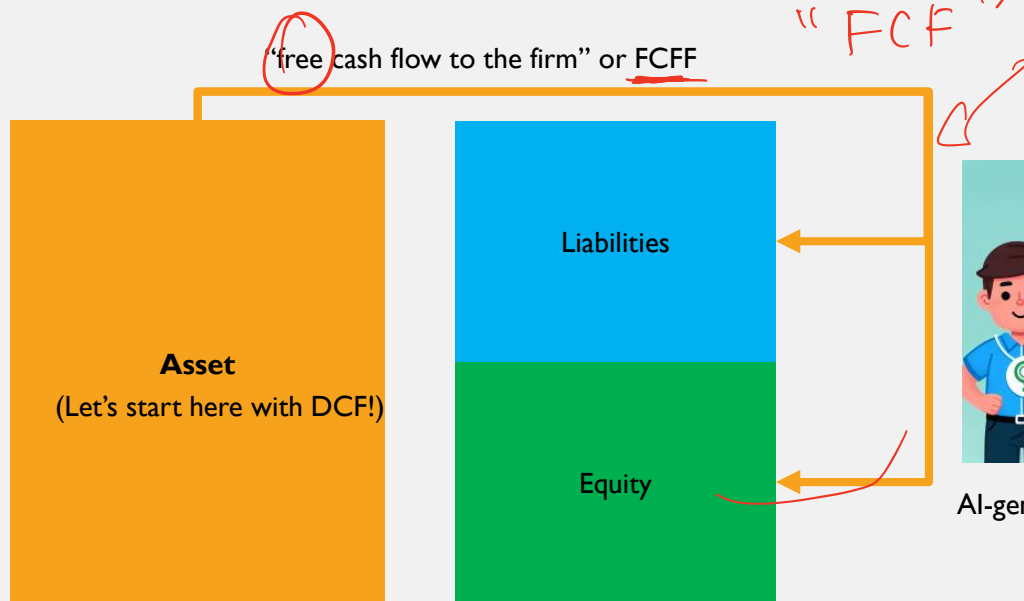
CAR Rental

100 CARs

non-op asset
mkt value.

0 0 0 0 0
PV $\rightarrow$

Start with the company's "assets", which generate FCFF



AI-generated image by copilot

Keke Song

7

Subject : Asset = Up Asset,
accrual

DCF
PV of CF.

Practice : non-op asset
DCF PV of CFs + Non-op asset
= Assets

Discounted Cash-Flow (DCF) Valuation

Use DCF to value a company's assets: basic idea

- Treat target firm's (operating) assets as a "box" that generates a stream of cash-flows.
- These cash flows from a firm's assets could have different names
 - "free cash-flows to the firm" } FCFF
 - "unlevered free cash flows" }
 - "cash flow from assets" }
- In this course, we often use FCFF to denote free cash-flows to the firm.
- The PV of a company's future FCFFs = the value of the firm's total assets (also called "total firm value")

2,3 year

5-10

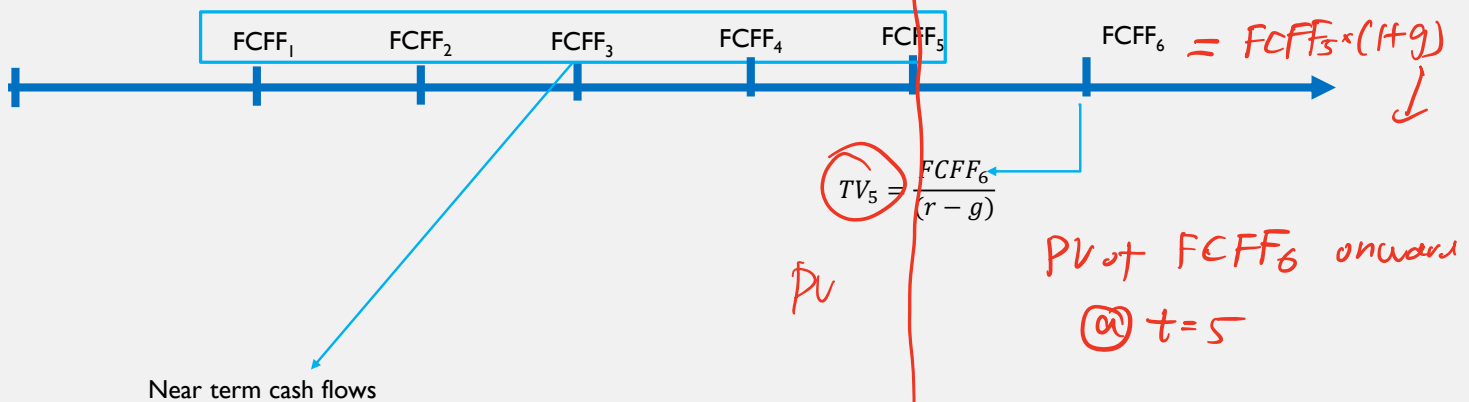
15 year



Example: Assume $T = 5$

g assumption

- Suppose we can forecast individual FCFFs to 5 periods from today. After that we assume the FCFF will grow at a constant rate, g .



K. Song

10

$$\underline{PV(FCFF)} = \frac{FCFF_1}{1+r} + \dots + \frac{FCFF_5}{(1+r)^5} + \frac{TV_5}{(1+r)^5}$$

Discounted Cash-Flow (DCF) Valuation

Valuing a company's assets: generic framework

$$TV_T = \frac{FCFF_{T+1}}{r - g} = \frac{FCFF_T \times (1 + g)}{r - g}$$

• Value of a company's assets = $\frac{FCFF_1}{(1+r)^1} + \frac{FCFF_2}{(1+r)^2} + \dots + \frac{FCFF_T}{(1+r)^T} + \frac{TV}{(1+r)^T}$

long-term perpetual growth rate

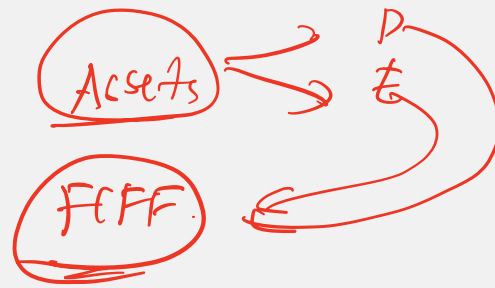
Discount rate

Cut-off year for the near-term (typically 5-10 years)

Discounted Cash-Flow (DCF) Valuation

Valuing a company's assets: discount rate choice

- Discussion: Which discount rate should we use in valuing the target's standalone value of total assets?
- The acquirer's cost of equity
- The acquirer's WACC
- The target's cost of equity
- ☒ The target's WACC



Discounted Cash-Flow (DCF) Valuation

Valuing a company's assets: discount rate choice

- Estimating cost of equity (r_e) using Capital Asset Pricing Model (CAPM):

$$r_e = r_f + \beta \times MRP$$

Inputs:

r_f = risk-free rate (proxied as yield on default-free government bond)

β = stock beta

MRP = market risk premium (proxied as average market return in excess of risk-free rate)

- Estimating pre-tax cost of debt (r_d):

- use yield(s) on firm's bonds (RATE function in Excel) → *bond available*
- if the firm does not have bonds, use bond yields of the firms with similar default risk (e.g., similar credit rating) *BBB*
- if neither of the above is available, use the interest rate on the company's most recent borrowing *AA*

↓ *F/S* → *Interest rate %* *Yield*

Keke Song

13

4

Interest Exp
Borrowing
↓

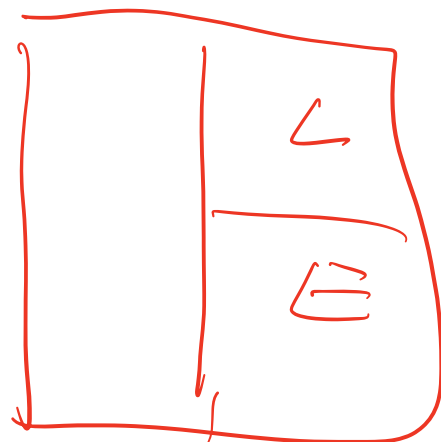
→ *\$5m* flow
100m
5%

Int Exp
2 months years

average (Borrowing t t-1)
2025 2024

51

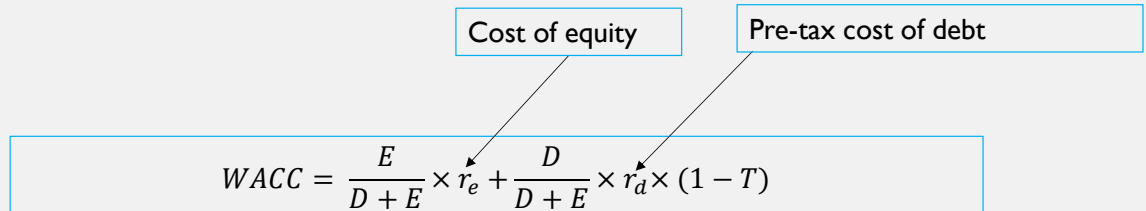
r_e > r_d



Discounted Cash-Flow (DCF) Valuation

Estimation of discount rate: generic framework

- Free cash flow to the firm (FCFF) represents cash available to satisfy all investors holding claims against the firm's assets.
- Thus, discount rate should represent opportunity costs of all investors (i.e., debt (D) and equity (E)) holding claims against firm's assets
- Weighted average cost of capital (WACC)


$$WACC = \frac{E}{D + E} \times r_e + \frac{D}{D + E} \times r_d \times (1 - T)$$

Summary of major steps in applying DCF in valuation

1. Estimate the WACC
2. Generate near-term pro-forma income statements
3. Calculate near-term future FCFFs → *Formulas*
4. Estimate the terminal value
5. Estimate the value of total assets or total firm value
6. Calculate the estimated value of equity

Keke Song



FCFFs: the cash flows generated by the company

$$FCFF = EBIT \times (1 - T_c) + D\&A - \Delta NWC - CapEx$$

$$EBIT \times (1 - T_c)$$

$$- (Capex - D\&A + \Delta NWC)$$

• ΔNWC = NWC in the current period minus NWC in the last period

• NWC = Current Assets - Current Liabilities

• Current Assets: Cash, Inventories, Accounts Receivables, etc.

• Current liabilities: Note that current liabilities do not include short-term debt

$$FCFF = EBIT \times (1 - T_c) - (CapEx - D\&A + \Delta NWC) \rightarrow \text{Net Reinvestment in NWC } \$30$$

$$\$200 - \$150$$

After-tax operating profit

$\$100m$

"Net Reinvestment"

$\$50$
→ Net Reinvestment in depreciable assets

• FCFF represents the cash flow available to distribute to all investors holding claims against the firm's assets

K. Song

16

$$= \$20m$$

$$\text{Net Reinvestment} \\ \$80m$$

Discounted Cash-Flow (DCF) Valuation

Estimation of near-term Free Cash Flows to Firm

- Use the spreadsheet “Module 3 & 4 in-class DCF exercise blank”
- Task 1: Estimate WACC of Air Thread in the spreadsheet:
- Task 2: Estimate near-term FCFF for Air Thread in the spreadsheet
 - Step 1: using the data in Excel spreadsheet, generate the Pro-Forma Income Statement for 2008 to 2012.
 - Step 2: generate the change in net working capital for 2008 to 2012.
 - Step 3: generate the projected capital expenditures.
 - Step 4: generate FFCF in the spreadsheet.

Discounted Cash-Flow (DCF) Valuation

Estimating value of a Company's assets: Task 1 and 2**

- Discount rate (WACC):

$$WACC = 0.7 \times 9.38\% + 0.3 \times 5.5\% \times (1 - 40\%) = 7.55\%$$

- Estimate near-term FCFF:

Period	0	1	2	3	4	5
FCFF		260.12	306.95	278.57	289.00	293.72

** refer to Excel spreadsheet calculations

Discounted Cash-Flow (DCF) Valuation

Terminal Value

- Long-term growth rate (g) should capture the stable rate of growth of FCFFs when the firm reaches its mature stage of the life cycle.
- Question: how would you estimate the long-term, perpetual growth rate
 1. long-term growth at the economy-wide level
 - average annual GDP growth or the average inflation rate
 - note: makes sense if the industry itself reaches a mature stage or the near-term period is fairly long
 2. long-term EBIT growth
 - estimate average **long-term EBIT growth rate**
 - go straight to the driving forces of growth!

Alternatively, you may choose to use the Long-term EBIT Growth Rate as the “g”

- Long-term EBIT growth = Net Reinvestment Rate * Return on Capital

$$= \frac{EBIT(1 - T_c)}{BV \text{ of Debt} + BV \text{ of Equity}}$$

$$= \frac{CapExp - Dep. \& Amort. + \Delta NWC}{EBIT(1 - T_c)}$$

- Note:

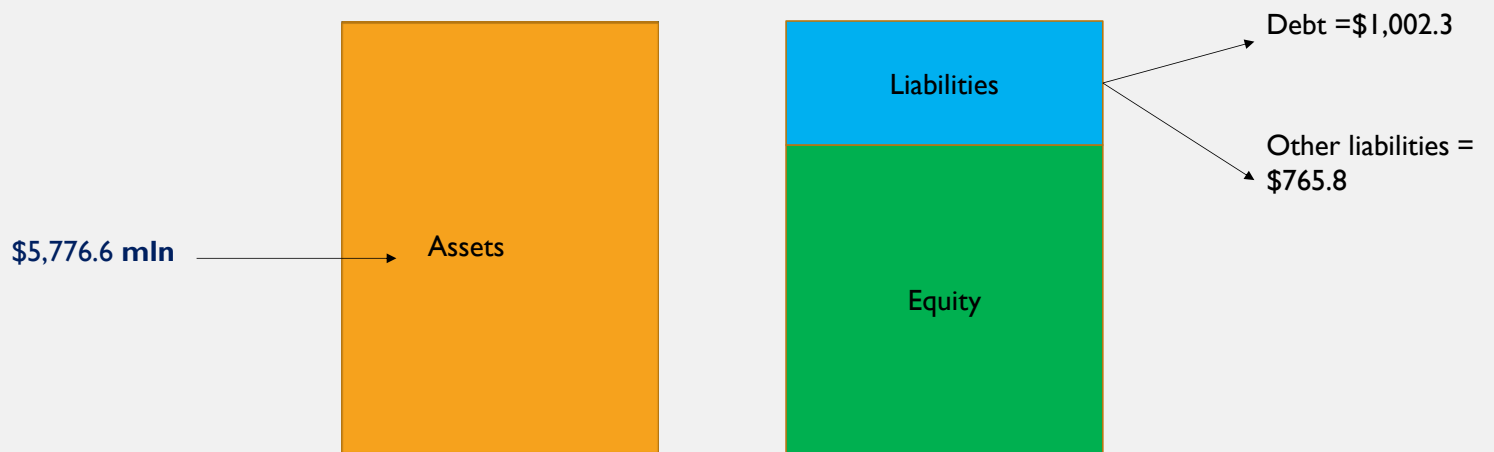
- Net Reinvestment Rate (NRR): proportion of reinvested capital in after-tax operating profits.
- Return on Capital (ROC): the return on reinvested capital.

Discounted Cash-Flow (DCF) Valuation

Estimating Sustainable Growth Rate: In-Class Practice

- Whose NRR and ROC to use?
 - Use the target company's historical average NRR and historical average ROC.
 - Or use the average NRR and ROC of comparable companies in the same industry.
- For example, if
 - The historical average NRR of the target company is **28.7%**.
 - The historical average ROC of the target company is **10.5%**.
 - The estimated long-term EBIT growth rate of the target company is $=28.7\% * 10.5\% =$

The Target Firm's Balance Sheet



The Value of the Target's Equity

Value of Equity	=	Assets	—	Liabilities
-----------------	---	--------	---	-------------

Value of Equity	=	\$5,776.6	—	(\$1,002.3 +\$765.8)
-----------------	---	-----------	---	-------------------------

Value of Equity	=	
-----------------	---	--

Share Value	=	<i>equity value ÷ shares outstanding</i>
-------------	---	--